



FPGA IMPLEMENTATION OF DC MOTOR SPEED AND DIRECTION CONTROL: A DECODING APPROACH

REVERSE ENGINEERING PROJECT REPORT

Submitted by

S.NISANTH	727623BEE015
S.THIRUVARUL	727623BEE065
G.T.KONGARASAN	727624BEE315

in partial fulfillment of the requirements for the degree of

BACHELOR OF ENGINEERING

IN

ELECTRICAL AND ELECTRONICS ENGINEERING

**DR. MAHALINGAM COLLEGE OF ENGINEERING AND
TECHNOLOGY**

AN AUTONOMOUS INSTITUTION

AFFILIATED TO ANNA UNIVERSITY

CHENNAI 600 025

NOVEMBER 2025

**DR. MAHALINGAM COLLEGE OF ENGINEERING AND
TECHNOLOGY, POLLACHI -642 003**

**(AN AUTONOMOUS INSTITUTION AFFILIATED TO
ANNA UNIVERSITY, CHENNAI - 600 025)**

BONAFIDE CERTIFICATE

Certified that this Reverse Engineering project report titled **“FPGA Implementation of DC Motor Speed and Direction Control: A Decoding Approach”** is the bonafide work of

S.NISANTH	727623BEE015
S.THIRUVARUL	727623BEE065
G.T.KONGARASAN	727624BEE315

who carried out the Reverse Engineering Project work under my supervision.

Dr. D.GANESH KUMAR M.E., Ph.D.,

PROFESSOR AND HEAD

Department of EEE
Dr.Mahalingam college of
Engineering and Technology
Pollachi- 642003

Dr. B.VINOTH KUMAR M.E., Ph.D.,

SUPERVISOR

Professor
Department of EEE
Dr.Mahalingam college of
Engineering and Technology
Pollachi- 642003

Submitted for the Autonomous End Semester Reverse Engineering Project work
Viva-Voce Examination held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

Dr. Mahalingam College of Engineering and Technology, Pollachi-642003

Academic Year: 2025 - 2026

TRL, SDG and Similarity Compliance Certificate

Project Title: FPGA Implementation of DC Motor Speed and Direction

Control: A Decoding Approach

Course Code: 23EEP501- Reverse Engineering Project

Department and Semester: EEE & V

Technology Readiness Level (TRL) of the Project	:	_____
Sustainability Development Goals (SDG)-Goal Name	:	_____
Similarity % from Turnitin Software	:	_____

we hereby declare that this Reverse Engineering Project report is original and complies with the institution's similarity guidelines.

S.No.	Names and Roll Numbers of Students	Signature of the Students
1	S. NISANTH -727623BEE015	
2	S.THIRUVARUL -727623BEE065	
3	G.T.KONGARASAN -727624BEE315	

Verified by

Name & Signature of the Guide

Head of the Department

Endorsed by

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

Recent progress in real-time digital control systems has been strongly driven by the performance and flexibility offered by modern FPGA architectures. In this project, signal decoding techniques are applied to reconstruct and implement Verilog based motor control logic on an FPGA platform. By decoding the original HDL behavior and translating it into a structured hardware design, the system is able to manage DC motor speed and direction with a high degree of precision. These decoding methods play a crucial role in generating stable PWM signals, maintaining consistent timing, and ensuring reliable direction control under varying operating conditions. When implemented on the Spartan-7 Boolean FPGA and interfaced with the L298N driver, the reconstructed logic enables smoother motor transitions, enhanced responsiveness, and predictable real-time performance. Overall, the work demonstrates how a decoding based FPGA approach can effectively bridge the gap between digital logic design and practical motor control applications.

ACKNOWLEDGEMENT

We wish to express our sincere thanks to all who have contributed to do this project through their support, encouragement and guidance.

We extend our gratitude to our management for having provided us with all facilities to build my project successfully. We express our sincere thanks to our honourable Secretary, Dr.S.V.Subramanian, MBA., Ph.D., for providing the required amenities.

We take this opportunity to express our deepest gratitude to our principal Dr.P.Govindasamy, Ph.D., who provide suitable environment to carry out the project.

We express our extreme gratefulness to Dr. D.GANESH KUMAR M.E., Ph.D., Professor and Head of the Department of EEE, for his constant support and encouragement.

We wish to express our deep sense of gratitude and thankfulness to my project guide Dr. B.VINOTH KUMAR, Professor, Department of EEE, for the valuable suggestion and guidance offered during the course of the project.

Finally, we are committed to place our heartfelt thanks to all those who had contributed directly and indirectly towards the success of the completion of this project.

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	IV
	LIST OF TABLES	VIII
	LIST OF FIGURES	IX
	LIST OF SYMBOLS AND ABBREVIATIONS	X
1	PRODUCT INTRODUCTION	1
1.2	PRODUCT DESCRIPTION	1
1.2.1	Overall System Architecture	2
1.3	OBJECTIVE	2
1.4	LITERATURE	3
2	METHODOLOGY AND PROCESS INVOLVED	4
2.1	TOOLS AND EQUIPMENT	
2.2	DECODING APPROACH	5
2.2.1	Decoding of Motor Control Module	5
2.3	RTL-LEVEL SCHEMATIC	6
2.4	SIMULATION AND LOGIC VERIFICATION	7
2.5	SYNTHESIS AND BITSTREAM GENERATION	7
2.6	HARDWARE INTEGRATION AND TESTING	8
3	FINDINGS & RESULTS	9
3.1	SYSTEM DESIGN	9
3.2	SUBSYSTEM DISSECTION ANALYSIS	10
3.3	INDIVIDUAL COMPONENT ANALYSIS	10

4	DISCUSSION & ANALYSIS	11
4.1	DESIGN ANALYSIS	11
4.2	EFFECTIVENESS OF DECODING APPROACH	12
4.3	CHALLENGES/ISSUES	12
5	CONCLUSION	13
5.1	KEY FINDINGS AND OUTCOMES OF THE PROJECT	13
5.2	SUGGESTIONS FOR FUTURE WORK	
5.3	APPLICATIONS	14
	APPENDIX 1	15
	APPENDIX 2	16
	REFERENCES	21

LIST OF TABLES

TABLE NO	TITLE	PAGE NO
2.1	Input and Output of Motor Control Module	6

LIST OF FIGURES

FIGURE NO	TITLE	PAGE NO
2.1	Block Diagram of Overall System	5
2.2	RTL Schematic of Control Module	7
2.3	Simulated Testbench Waveform	7
2.4	Bitstream Generation	8
2.5	Pin Mapping & Hardware Integration	8
3.1	Overall System Design	9
4.1	Different Duty Cycle	11

LIST OF SYMBOLS AND ABBREVIATIONS

FPGA	-	Field Programmable Gate Array
PWM	-	Pulse Width Modulation
DC	-	Direct Current
HDL	-	Hardware Description Language
VHDL	-	VHSIC Hardware Description Language
IoT	-	Internet of Things
DSO	-	Digital Storage Oscilloscope
V	-	Volts
PI	-	Proportional-Integral
PID	-	Proportional-Integral-Derivative

CHAPTER 1

PRODUCT INTRODUCTION

This chapter focuses the significance of FPGA based motor control systems and the use of decoding strategies to regulate motor performance. It also presents a concise review of the literature relevant to the current study.

1.2 PRODUCT DESCRIPTION

The unique behavior of a DC motor comes from its ability to convert electrical energy into rotational motion with a balance that lies between the rigid characteristics of mechanical systems and the dynamic responsiveness of electronic systems. Unlike static mechanical components, a DC motor does not maintain a fixed state; it adjusts its speed based on the applied voltage and changes direction when the supply polarity is reversed.

Integrating the motor with an FPGA based controller further enhances its functionality. The FPGA enables precise, high speed decoding of control inputs and the generation of finely tuned PWM signals, allowing the motor to react instantly to user commands or automated control logic. Because the FPGA operates with deterministic timing and parallel processing, it ensures stable performance, even when the motor is subjected to load disturbances or sudden variations in operating conditions. Overall, this combination results in a highly responsive, reliable, and efficient motor control system suitable for real-time applications.

1.2.1 Overall System Architecture

The overall system architecture integrates the reverse engineered motor control logic with an FPGA based implementation capable of generating precise PWM and direction signals. At the core of the system is the Spartan-7 Boolean FPGA, which decodes user inputs or reconstructed control patterns and translates them into the appropriate duty cycle values and direction commands. These signals are then fed into the L298N motor driver, which provides the necessary current and voltage amplification to drive the 12 V DC motor. The FPGA handles all real-time signal processing, including PWM timing, direction switching, and safety control, ensuring deterministic behavior. The architecture follows a modular structure consisting of input decoding, PWM generation, motor driver interfacing, and optional feedback monitoring. This design not only replicates the functional behavior of the original system but also enhances flexibility, allowing easy modification and extension through reprogrammable logic

1.3 OBJECTIVE

The main objective of this project is to develop, analyse, and validate a PWM based motor control system using Verilog HDL on an FPGA platform. The work focuses on generating accurate PWM signals for regulating motor speed, implementing reliable direction control logic, and ensuring that the Verilog description behaves consistently when deployed on hardware. Another key goal is to simulate and test the design thoroughly before implementation, allowing performance verification at each stage. By comparing simulation results with the real-time response of the DC motor, the project aims to establish a clear link between HDL code, FPGA execution, and actual motor behavior.

1.4 LITERATURE

FPGAs have gained significant attention in motor control applications due to their deterministic timing, high parallelism, and ability to implement custom hardware logic suited for real-time systems. Several studies highlight the advantages of generating PWM signals directly in HDL, showing that FPGA based PWM provides finer resolution, and more stable speed control compared to microcontroller driven methods. Researchers have also demonstrated that FPGA implementations allow flexible duty cycle modulation and reliable direction switching, making them well suited for DC motor applications where quick response and precision are essential.

Closed loop motor control techniques implemented on FPGAs are also widely discussed in the literature. Many works show successful use of PI or PID controllers written in Verilog or VHDL to maintain constant motor speed under varying load conditions. These studies emphasize the importance of synchronous sampling, fixed point arithmetic, and proper tuning to ensure stable performance. Practical implementations frequently combine PWM generation with feedback signals from encoders or tachometers to achieve accurate speed regulation.

Several projects based studies have used FPGA boards alongside drivers such as the L298N for controlling small DC motors, focusing on safe interfacing, direction logic, and hardware verification. Overall, the literature supports the effectiveness of FPGA based PWM and control logic, forming a strong foundation for developing a decoding based DC motor control system.

CHAPTER 2

METHODOLOGY AND PROCESS INVOLVED

A detailed decoding procedure adopted for this project is presented in this chapter. In addition, the digital design principles involved in generating PWM signals and implementing FPGA based motor control logic are also discussed.

2.1 TOOLS AND EQUIPMENT

HARDWARE

- ❖ Xilinx Spartan-7 Boolean FPGA Board
- ❖ 12V DC Motor (AM37B5203012116)
- ❖ L298N Motor Driver Module
- ❖ DSO for PWM and duty cycle visualization
- ❖ Regulated Power Supply and Connecting Wires

SOFTWARE

- ❖ Verilog HDL
- ❖ Xilinx Vivado for HDL coding, simulation, synthesis, and implementation

2.2 DECODING APPROACH

The decoding approach in this project focuses on understanding the Verilog code used to control the speed and direction of the DC motor. The original HDL design was examined line by line to identify how duty cycle values were assigned, how the PWM counter operated, and how direction signals were generated. By interpreting the logic behind the case statements, combinational assignments, and sequential always blocks, the functional behavior of the system was fully decoded.

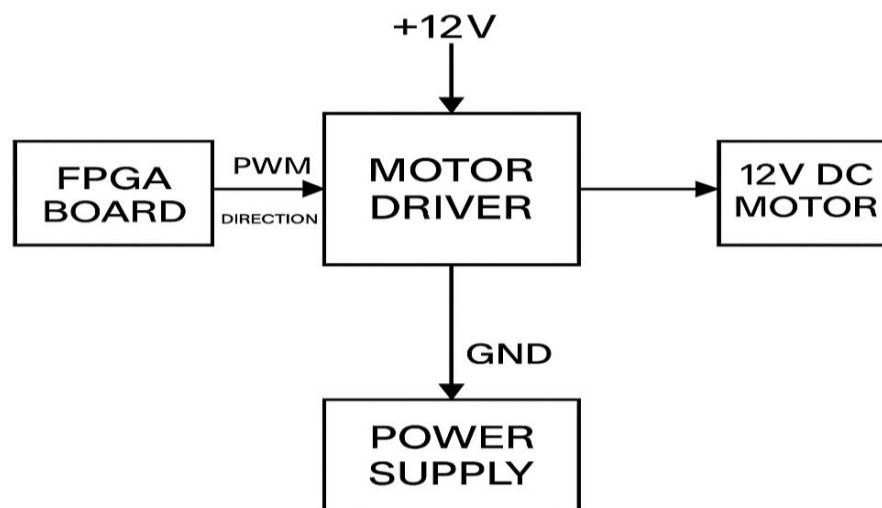


Figure 2.1 Block Diagram of Overall System

2.2.1 Decoding of Motor Control Module

In this project, the motor-control logic was decoded and re-implemented using behavioral Verilog. The aim was to understand how PWM generation and direction control were defined in the original code and reproduce that functionality on the FPGA. The design was broken down into three blocks: duty cycle selection, direction assignment, and PWM generation. The *value* input set predefined duty levels through a behavioral *case* structure, while the *dir* input controlled *in1* and *in2* for motor rotation. A rising edge counter compared against *temp* produced the PWM output. This decoding clarified both the combinational

logic for duty and direction and the sequential logic for PWM, enabling accurate FPGA implementation on the Spartan-7 Boolean board.

Table 2.1 Input and Output of Motor Control Module

Signal	Type	Description
clk	input	System clock signal used to drive the PWM timing (counter increments on every clock pulse).
rst	input	Active high reset signal; when high, it clears the counter and PWM output.
value [1:0]	input	Selects the PWM duty cycle determines how long the signal stays HIGH in each period.
dir	input	Direction control signal decides motor rotation direction (forward or reverse).
pwm_out	output	PWM signal controlling motor speed
motor_out1	output	Logic signal for motor direction
motor_out2	output	Logic signal for motor direction

2.3 RTL-LEVEL SCHEMATIC

The schematic shows the synthesized hardware structure derived from the decoded Verilog design. A mux selects one of four duty cycle values based on *value*. The *dir* input drives output muxes to set in1 and in2.

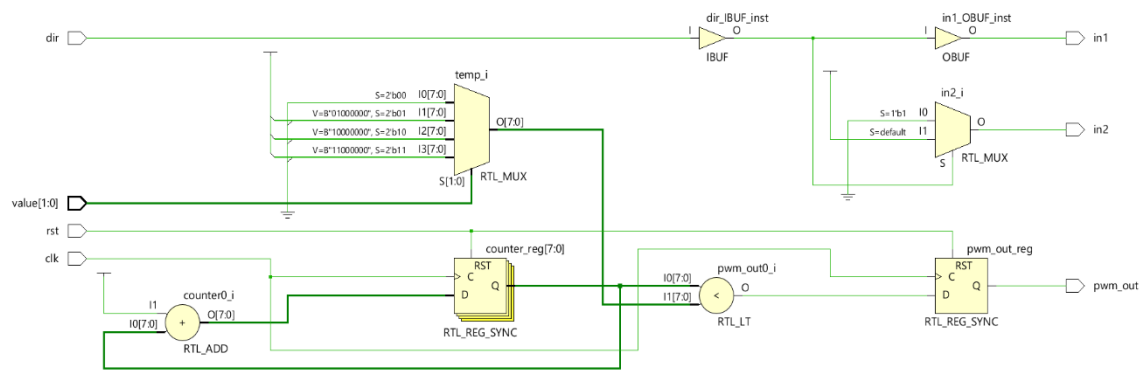


Figure 2.2 RTL Schematic of Control Module

2.4 SIMULATION AND LOGIC VERIFICATION

The Verilog design was simulated in Vivado 2019.2 to verify the behavior of the PWM generator and direction control logic. Testbench inputs were applied to observe how the duty cycle values changed for each selection and whether the PWM waveform matched the expected pattern. The simulation also confirmed that the direction signals (*in1* and *in2*) correctly toggled based on the *dir* input before moving to hardware implementation.

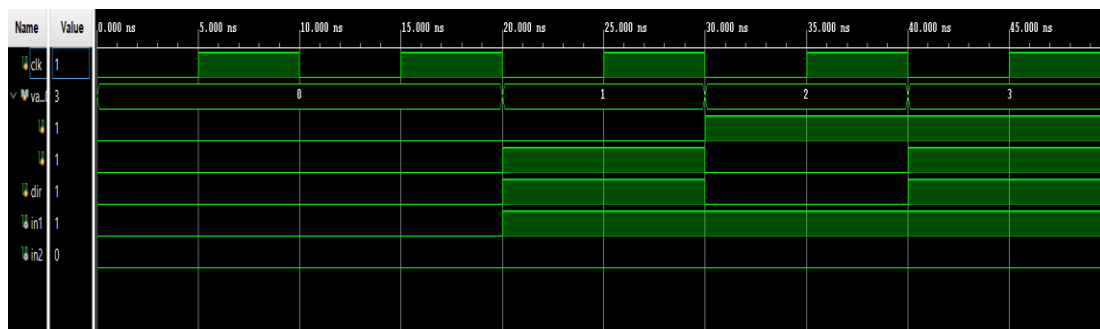


Figure 2.3 Simulated Testbench Waveform

2.5 SYNTHESIS AND BITSTREAM GENERATION

The decoded design was synthesized to map the Verilog logic onto the Spartan-7 Boolean FPGA board. Vivado synthesis and implementation steps optimized the PWM and direction control blocks, followed by bitstream generation for hardware deployment.

```

sd.xdc
D:/Digital Vlsi/FPGA prototyping/boolean_board_new/boolean_board_new.srsc/constrs_1/new/sd.xdc

1 set_property -dict {PACKAGE_PIN F14 IOSTANDARD LVCMOS33} [get_ports {clk} ]
2 set_property -dict {PACKAGE_PIN K1 IOSTANDARD LVCMOS33} [get_ports {rst}]
3 set_property -dict {PACKAGE_PIN V2 IOSTANDARD LVCMOS33} [get_ports {value[0]}]
4 set_property -dict {PACKAGE_PIN U2 IOSTANDARD LVCMOS33} [get_ports {value[1]}]
5 set_property -dict {PACKAGE_PIN U1 IOSTANDARD LVCMOS33} [get_ports {dir}]
6 set_property -dict {PACKAGE_PIN B13 IOSTANDARD LVCMOS33} [get_ports {in1}]
7 set_property -dict {PACKAGE_PIN A13 IOSTANDARD LVCMOS33} [get_ports {in2}]
8 set_property -dict {PACKAGE_PIN B14 IOSTANDARD LVCMOS33} [get_ports {pwm_out}]
9
10

```

Figure 2.4 Bitstream Generation

2.6 HARDWARE INTEGRATION AND TESTING

The programmed FPGA board was connected to the L298N motor driver, with PWM, *in1*, and *in2* lines interfaced directly. Power was supplied to the 12 V DC motor, and the setup was tested to ensure correct speed and direction control.

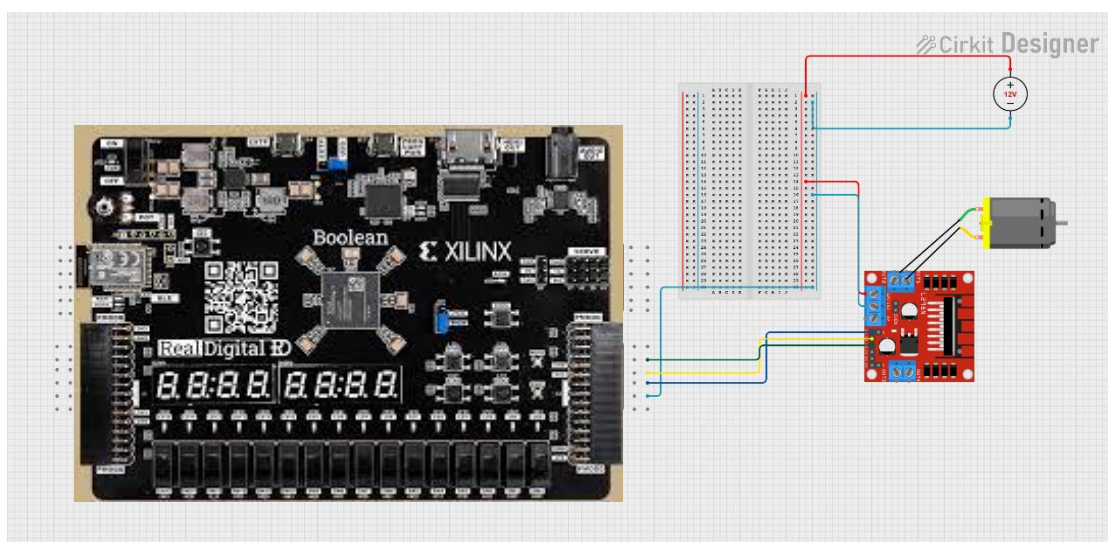


Figure 2.5 Pin Mapping & Hardware Integration

CHAPTER 3

FINDINGS & RESULTS

The results obtained from the decoding the FPGA based motor control system are presented in this chapter. They highlight the reconstructed Verilog behavior, hardware performance, PWM accuracy, and overall effectiveness of the implemented decoding approach.

3.1 SYSTEM DESIGN

The system wide analysis shows that the FPGA successfully replicated the decoded Verilog control logic for both PWM generation and direction control. The implemented design maintained stable operation across all tested duty cycle levels 0%, 25%, 50%, and 75%. The reconstructed logic produced clean PWM waveforms with consistent timing when verified using the DSO

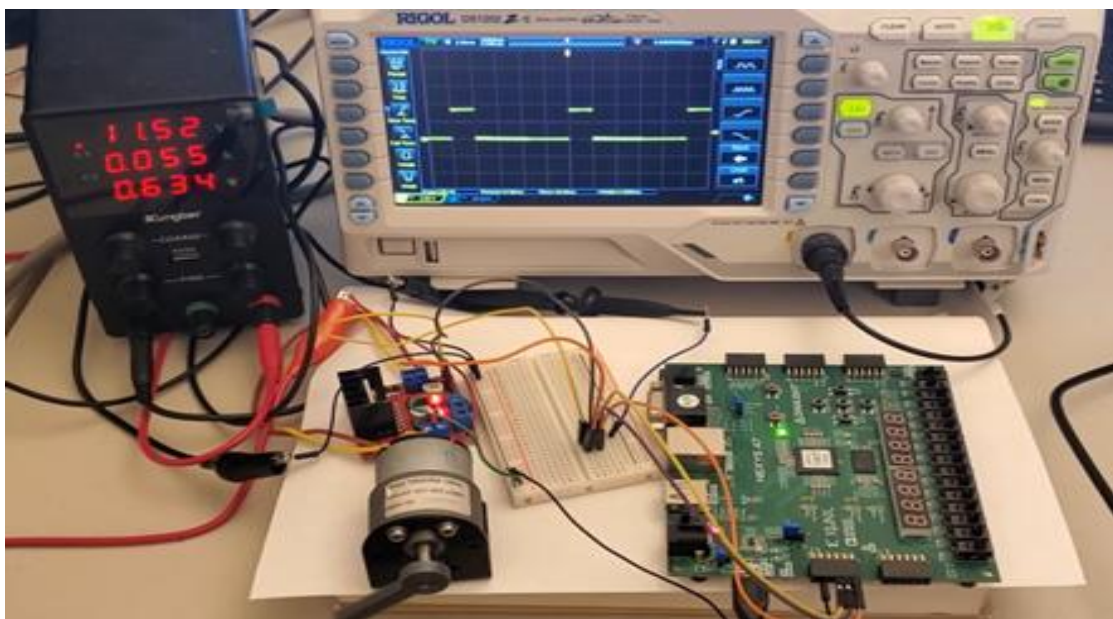


Figure 3.1 Overall System Design

3.2 SUBSYSTEM DISSECTION ANALYSIS

Subsystem level analysis focused on verifying each functional block of the Verilog design.

- ❖ PWM Subsystem: The counter based PWM generator produced correct duty cycle widths for all input combinations.
- ❖ Direction Control Subsystem: The decoding of the *dir* signal was validated by monitoring *in1* and *in2* pins. Forward and reverse operations were triggered cleanly, with safe and immediate switching.
- ❖ Decoding Logic: The behavioral case structure used for speed levels worked as intended, accurately mapping binary inputs to predefined speed values.

3.3 INDIVIDUAL COMPONENT ANALYSIS

Individual component behavior was evaluated to understand their roles within the overall system.

- ❖ Spartan-7 Boolean FPGA Board: Successfully synthesized and executed the decoded Verilog design with low resource usage and stable timing performance.
- ❖ L298N Motor Driver: Performed well under the PWM switching frequency used, providing adequate current to the motor without overheating.
- ❖ 12 V DC Motor: Responded proportionally to changes in duty cycle, showing smooth speed transitions and correct direction switching.
- ❖ DSO: Verified PWM pulse width, frequency stability, and switching integrity, confirming accurate reproduction of decoded logic.

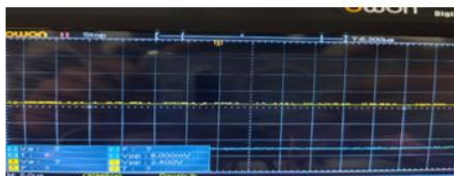
CHAPTER 4

DISCUSSION AND ANALYSIS

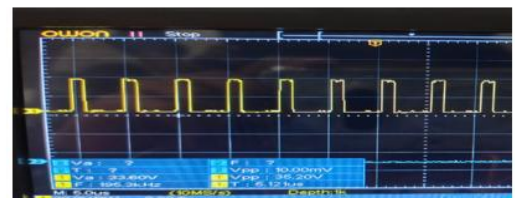
4.1 DESIGN ANALYSIS

The design analysis shows that the system architecture consisting of input decoding, PWM generation, and direction control was implemented efficiently using behavioral Verilog. The counter based PWM logic produced clean switching edges, and the FPGA handled the updates at each clock cycle without timing violations. The L298N driver integrated seamlessly with the FPGA outputs, enabling stable motor operation. The chosen duty cycle values (0%, 25%, 50%, 75%) provided clear speed variation, and direction signals toggled accurately during hardware testing. The compact and modular structure of the design also allows easy scalability for future improvements.

0% Duty Cycle



25% Duty Cycle



50% Duty Cycle



75% Duty Cycle

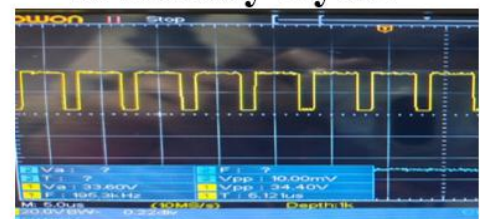


Figure 4.1 Different Duty Cycle

4.2 EFFECTIVENESS OF DECODING APPROACH

The effectiveness of the decoding approach was evaluated by examining how accurately the reconstructed Verilog logic reproduced the original motor control behavior. The decoding process successfully identified the functional roles of each block duty cycle mapping, direction handling, and PWM timing and translated them into a clear and implementable FPGA structure. When deployed on the Spartan-7 FPGA, the generated PWM waveforms matched the expected duty cycle values, and direction signals responded reliably to all input conditions. Hardware tests with the L298N driver further confirmed that the decoded logic behaved consistently, enabling smooth speed transitions and correct directional movement of the motor. The close alignment between the decoded HDL behavior, simulated outputs, and real-time motor performance demonstrates that the decoding methodology was effective and allowed accurate reconstruction of the original design.

4.3 CHALLENGES / ISSUES

During the development and testing phase, several challenges were encountered:

- ❖ **Voltage Level Mismatch:** Ensuring safe interfacing between the FPGA's 3.3V logic and the L298N required proper design considerations.
- ❖ **Motor Load Variations:** Under sudden load changes, the motor speed exhibited minor fluctuations, highlighting the need for feedback based control in future upgrades.
- ❖ **DSO Measurement Sensitivity:** Minor distortions appeared when probing the PWM waveform at high speed, requiring stable probe grounding

CHAPTER 5

CONCLUSION

This project successfully reverse engineered the Verilog motor control logic and implemented it on an FPGA using a decoding based approach. The PWM and direction control functions were reconstructed through behavioral modelling and validated through hardware testing. The Spartan-7 Boolean FPGA, combined with the L298N driver and 12 V DC motor, delivered stable real-time performance, demonstrating the effectiveness of the decoded HDL logic in controlling physical motor behavior.

5.1 KEY FINDINGS AND OUTCOMES OF THE PROJECT

- ❖ The Verilog design was successfully decoded and implemented using behavioral modelling.
- ❖ PWM signals generated by the FPGA matched the expected duty cycle and timing patterns.
- ❖ The L298N driver responded accurately to direction and PWM inputs.
- ❖ Hardware tests confirmed correct motor operation under all tested conditions.

5.2 SUGGESTIONS FOR FUTURE WORK

- ❖ Add Speed Feedback: Integrating a rotary encoder or hall sensor would enable closed loop speed control.
- ❖ Implement PID Controller: A digital PI/PID controller running on the FPGA can further stabilise speed under load variations
- ❖ Increase PWM Resolution: Higher bit PWM can produce smoother motor speed transitions.

5.3 APPLICATIONS

- ❖ It is used in robotic arms and automated actuation systems that require precise and responsive motor control.
- ❖ It is employed in conveyor belt speed regulation within industrial automation setups to ensure smooth and stable material handling.
- ❖ It is used in smart home and IoT based motorized devices, enabling controlled motion in compact and intelligent embedded systems.
- ❖ It is employed in automated doors, lifts, and access control mechanisms where reliable direction and speed control are essential.
- ❖ It is used in general purpose electromechanical systems that rely on efficient and flexible motor control for consistent operation.

APPENDIX 1

VERILOG CODE

Design

```

`timescale 1ns/1ps

module pwm(
    input clk,
    input rst,
    input [1:0] value, // select duty cycle
    input dir,         // direction: 0 or 1
    output reg pwm_out, // connect to ENA
    output reg in1,     // connect to IN1
    output reg in2      // connect to IN2
);
    reg [7:0] counter;
    reg [7:0] temp;

//DUTY CYCLE VARIATION

    always @(*) begin
        case(value)
            2'b00: temp = 8'd0;    // 0% MOTOR
OFF        2'b01: temp = 8'd64;   // 25%
            2'b10: temp = 8'd128;  // 50%
            2'b11: temp = 8'd192;  // 75%
        endcase
    end

// MOTOR DIRECTION
    always @(*) begin
        if(dir) begin
            in1 = 1'b1;
            in2 = 1'b0;
        end else begin
            in1 = 1'b0;
            in2 = 1'b1;
        end
    end

//PWM GENERATION
    always @(posedge clk) begin
        if(rst) begin
            counter <= 8'd0;
            pwm_out <= 1'b0;
        end else begin
            counter <= counter + 1'b1;
            if(counter < temp)
                pwm_out <= 1'b1;
            else
                pwm_out <= 1'b0;
        end
    end
endmodule

```

Testbench

```

`timescale 1ns/1ps
module pwm_tb;
    reg clk;
    reg rst;
    reg [1:0] value;
    reg dir;
    wire in1, in2;
    pwm uut (
        .clk(clk),
        .rst(rst),
        .value(value),
        .dir(dir),
        .in1(in1),
        .in2(in2)
    );
    initial begin
        clk = 0;
        forever #5
            clk = ~clk;
    end
    initial begin

        rst = 1;
        value = 2'b00;
        dir = 0;
        #10;
        rst = 0;
        #10;
        value = 2'b01;
        #10;
        dir = 1;
        #10;
        value = 2'b10;
        dir = 0;
        #10;
        dir = 1;
        #10;
        value = 2'b11; dir = 0;
        #10;
        dir = 1;
        #10;
        $stop;
    end
endmodule

```

APPENDIX 2

COMPONENT DATASHEETS

DC Geared Motor – 12V (10 to 1000 RPM)

Key Features

- DC supply: 4 to 12V
- Total length: 46mm
- Motor diameter: 36mm
- Motor length: 25mm
- Brush type: Precious metal
- Gear head diameter: 37mm
- Gear head length: 21mm
- Output shaft: Centred
- Shaft diameter: 6mm
- Shaft length: 22mm
- Gear assembly: Spur



Weight	50g
Dimensions	6 x 6 x 1 cm
RPM Options	10,30,60,100,150,200,300,500,1000 RPM
Gear Head Length	21mm
Output Shaft	Centred
Brush Type	Precious Metal

MOTOR DRIVER

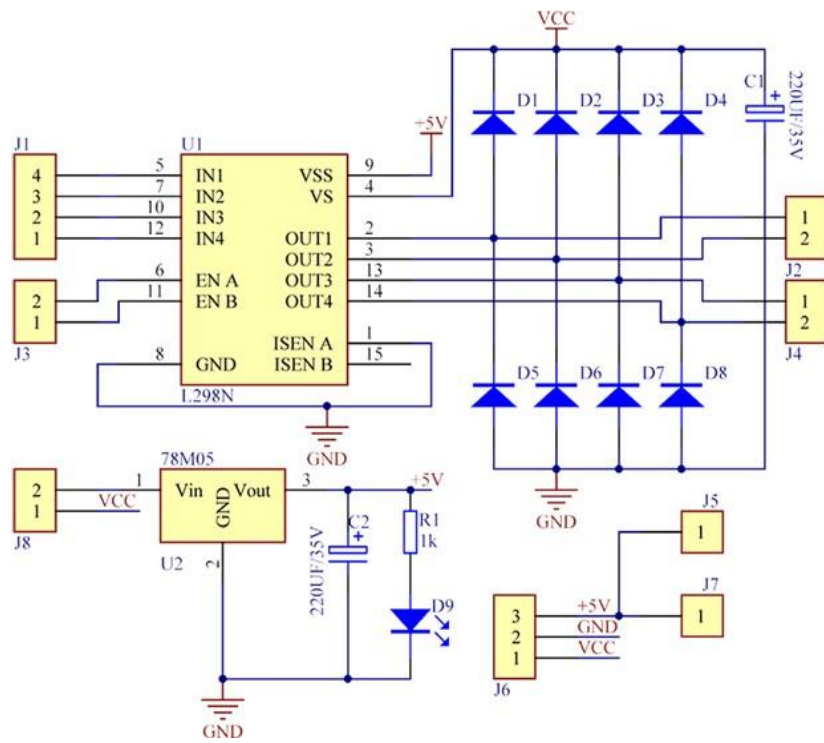
User Guide

This dual bidirectional motor driver, is based on the very popular L298 Dual H-Bridge Motor Driver Integrated Circuit. The circuit will allow you to easily and independently control two motors of up to 2A each in both directions. It is ideal for robotic applications and well suited for connection to a microcontroller requiring just a couple of control lines per motor. It can also be interfaced with simple manual switches, TTL logic gates, relays, etc. This board equipped with power LED indicators, on-board +5V regulator and protection diodes.

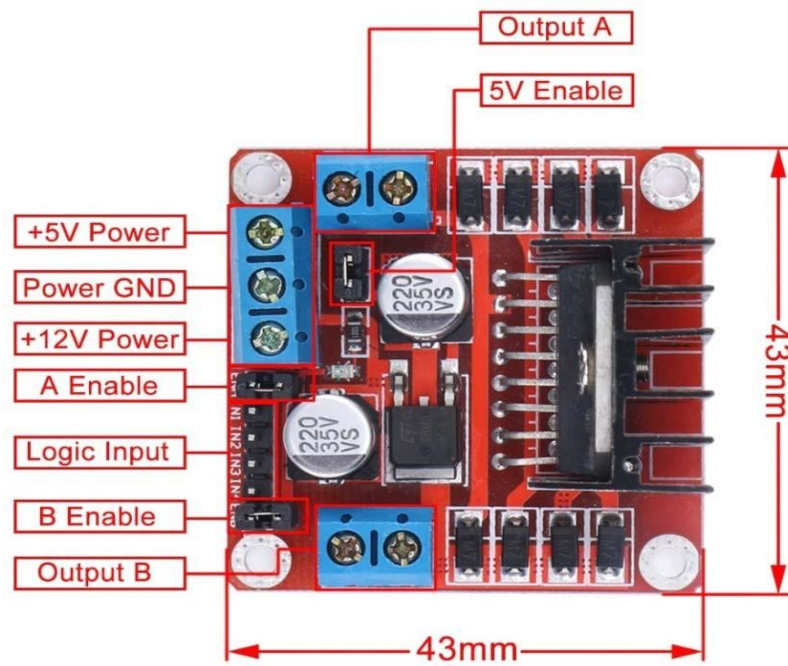
Brief Data:

- ❖ Input Voltage: 3.2V~40Vdc.
- ❖ Driver: L298N Dual H Bridge DC Motor Driver
- ❖ Power Supply: DC 5 V - 35 V
- ❖ Peak current: 2 Amp
- ❖ Operating current range: 0 ~ 36mA
- ❖ Control signal input voltage range :
- ❖ Low: $-0.3V \leq V_{in} \leq 1.5V$.
- ❖ High: $2.3V \leq V_{in} \leq V_{ss}$.
- ❖ Enable signal input voltage range :
 - Low: $-0.3 \leq V_{in} \leq 1.5V$ (control signal is invalid).
 - High: $2.3V \leq V_{in} \leq V_{ss}$ (control signal active).
- ❖ Max power consumption: 20W (when the temperature $T = 75\text{ }^{\circ}\text{C}$).
- ❖ Storage temperature: $-25\text{ }^{\circ}\text{C} \sim +130\text{ }^{\circ}\text{C}$.
- ❖ On-board +5V regulated Output supply (supply to controller board i.e. Arduino).
- ❖ Size: 3.4cm x 4.3cm x 2.7cm

Board Dimension & Pins Function:



Schematic Diagram:



FPGA BOARD

Boolean Board

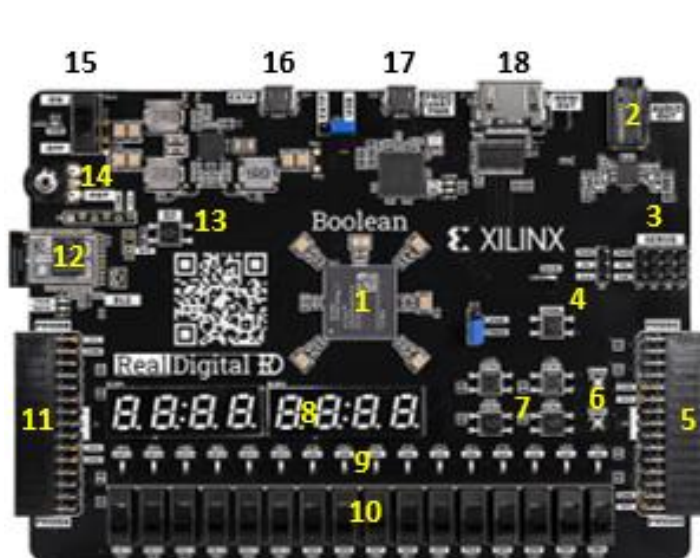
The Boolean board is a configurable digital logic system designed specifically for teaching and learning digital logic design. The board is centered on a Xilinx Spartan-7 FPGA (XC7S50) that can be configured into a virtually unlimited number of hardware circuits. The Spartan-7 FPGA includes:

- ❖ more than 8,000 FPGA slices, each containing four 6-input look-up tables and 8 flip-flops, equivalent to 52K logic cells
- ❖ 120 DSP blocks, each with dual 24-bit adders, a 2's complement multiplier, and a 48-bit accumulator
- ❖ 337Kbytes of high-speed, dual port RAM
- ❖ Five clock management tiles that can generate a wide variety of clock signals from a single outside source
- ❖ A 12-bit, 1MSPS analog-to-digital converter
- ❖ And many other circuits and functions.

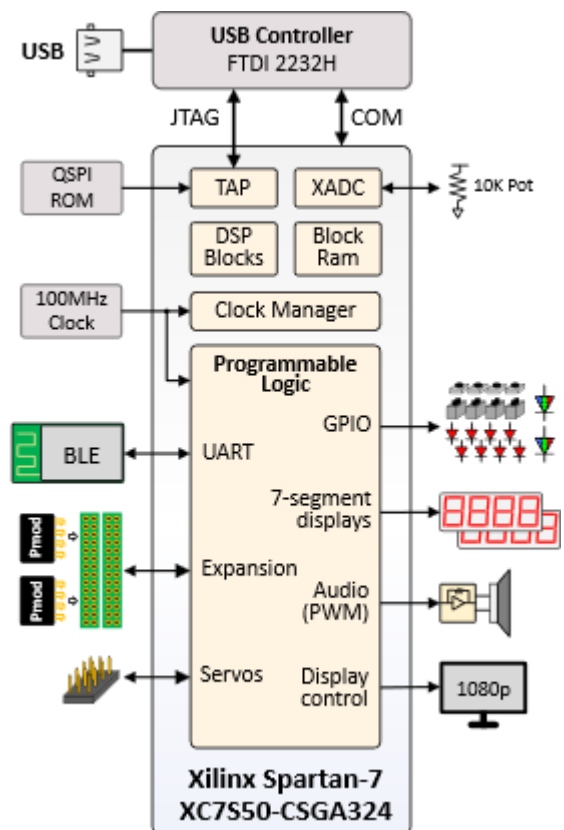
The Boolean board surrounds the FPGA chip with several peripheral devices, including push buttons, slide switches, discrete and RGB LEDs, an 8-digit seven-segment display, audio and video outputs and an optional Bluetooth radio – everything needed to implement a wide variety of digital circuits and systems without the need for any other components. Two 30-pin expansion connectors, both capable of driving two Pmods as well as custom peripheral boards, provide easy access to an additional 44 FPGA I/O pins.

Highlights

- ❖ Spartan 7 XC7S50 FPGA
- ❖ USB port for power & programming
- ❖ On-board Bluetooth Low Energy radio
- ❖ HDMI -1080p
- ❖ On-board ADC
- ❖ QSPI for storing configurations and user data
- ❖ Four servo connectors
- ❖ An 8-digit seven-segment display
- ❖ 5 push-buttons, 16 slide switches, 16 LEDs, 2 RGB LEDs
- ❖ Four Pmod-compatible expansion connectors



1. Spartan-7 XC7S50-CSGA324 FPGA	10. 16 slide switches
2. Audio output	11. Expansion connector (30-pin)
3. Servo motor connectors (x4)	12. BLE radio module
4. FPGA Program button (reset)	13. BLE reset button
5. Expansion connector (30-pin)	14. 10K Potentiometer (connected to ADC)
6. 2 RGB LEDs	15. Power switch
7. 4 pushbuttons	16. Accessory power connector
8. 8-digit seven-segment display	17. Main USB (power, programming, UART)
9. 16 discrete LEDs	18. HDMI source (up to 1080p)



REFERENCES

1. Gupta, A & Verma, R 2013, 'Decoding-based PWM Speed Control of DC Motor using FPGA', *IEEE International Conference on Computational Intelligence and Communication Networks*, DOI: 10.1109/CICN.2013.188.
2. Mano, M.M & Ciletti, M.D 2017, *Digital Design with an Introduction to the Verilog HDL*, 6th edn, Pearson Education, London.
3. Xilinx Inc. 2019, *Vivado Design Suite User Guide*, Available from: <https://www.xilinx.com/support/documentation-navigation/design-hubs/dh0002-vivado-design-hub.html> [Accessed: 04 Feb 2025].
4. STMicroelectronics 2018, *L298N Motor Driver Datasheet*, Available from: <https://www.st.com/en/motor-drivers/l298.html> [Accessed: 04 Feb 2025].
5. Kumar, S & Singh, M 2013, 'PWM Based Speed Control of DC Motor using FPGA', *IEEE International Conference on Computational Intelligence and Communication Networks*, DOI:10.1109/CICN.2013.188.
6. Palani, R & Sundaram, P 2014, 'FPGA Implementation of High-Precision PWM for Motor Control Applications', *International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering*, vol. 3, no. 5.
7. Prakash, A & Rao, D 2016, 'Design of FPGA-based DC Motor Speed Controller Using Verilog HDL', *International Conference on Emerging Trends in Engineering & Technology*.
8. El-Hameed, A & Abdelhady, S 2012, 'FPGA-Based PID Controller for DC Motor Speed Control', *International Journal of Control, Automation and Systems*, vol. 10, no. 4.
9. Rao, S & Devi, K 2018, 'Analysis of H-Bridge Drivers for DC Motor Control', *Journal of Power Electronics and Drives*, vol. 6, no. 2.
10. Boolean Labs 2020, *Boolean FPGA Development Board – Technical Reference Manual*, Boolean Embedded Systems Pvt. Ltd.